

Enthusiast COURSE

PHASE : (RBT)

TARGET : PRE-MEDICAL 2025

Test Type : MAJOR

Test Pattern : NEET (UG)

TEST DATE : 01-04-2025

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	2	4	1	4	1	2	1	3	2	4	1	2	2	3	4	4	1	3	3	3	2	1	3	3	2	1	2	1	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	2	2	2	3	3	3	2	3	1	2	1	4	3	2	3	2	1	4	4	3	2	4	2	3	1	4	4	3	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	4	4	2	3	4	4	2	1	1	2	1	3	1	1	2	2	4	2	2	4	1	3	4	4	3	4	3	4	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	2	3	4	4	3	2	2	4	2	1	1	1	3	2	2	2	2	2	2	3	3	4	1	4	2	4	4	2	1
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	4	2	1	2	1	3	4	3	4	4	1	2	1	1	1	3	2	1	4	3	1	1	3	3	4	3	4	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	4	4	1	1	2	1	2	4	3	3	1	2	3	2	2	1	4	4	2	2	1	3	1	1	3	4	2	1	2

HINT - SHEET

1. **Ans (3)**

$$\mu = \tan \lambda \quad \lambda = \text{Angle of friction}$$

2. **Ans (2)**

$$F_{av} = \frac{m(V_f - V_i)}{\Delta t} = \frac{250 \times 3.8}{1000 \times 0.02} = 47.5 \text{ N}$$

3. **Ans (4)**

$$C = \sqrt{N^2 + f^2}$$

4. **Ans (1)**

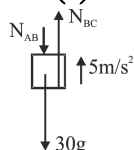
For (1) : Net force on B = $ma = (30)(5) = 150 \text{ N}$

For (2) : FBD of A



$$N_{AB} - 20g = 20(5) \Rightarrow N_{AB} = 300 \text{ N}$$

For (3) : FBD of B



$$N_{BC} - N_{AB} - 30g = 30(5) \Rightarrow N_{BC} = 750 \text{ N}$$

For (4) : For system of A + B + C :

$$N_C - 100g = 100(5) \Rightarrow N_C = 1500 \text{ N}$$

5. **Ans (4)**

Acceleration down a rough inclined plane is given by,
 $a = g(\sin \theta - \mu \cos \theta)$ and this is less than g .

6. **Ans (1)**

$$mg(1) = \frac{1}{2} \mu u^2 \Rightarrow u^2 = 2g$$

$$\therefore v^2 = u^2 + 2aS$$

$$\Rightarrow 0 = 2g - 2\mu g \times 3$$

$$\therefore \mu = \frac{1}{3}$$

7. **Ans (2)**

$$w = Fd \cos \theta$$

$$\text{when } \vec{F} \perp \vec{d} \Rightarrow \theta = 90^\circ \Rightarrow w = 0$$

$$\text{when } \theta > 90^\circ \Rightarrow \cos \theta = \ominus \text{ive} \Rightarrow w = \ominus \text{ive}$$

8. **Ans (1)**

$$u = 5 \text{ m/s}, v = 1/10 \text{ m/s}, a = -\frac{10^{-3}}{0.01} = -\frac{1}{10} \text{ m/s}^2$$

$$\Rightarrow t = \frac{v - u}{a} = \frac{\frac{1}{10} - 5}{-\frac{1}{10}} = 49 \text{ sec}$$

9. Ans (3)

$$\vec{F}_1 = (3N)\hat{i} - (4N)\hat{j}$$

This means (3N) along + x axis and (4N) along negative side of y axis. Hence vector 5 represents it.

$$\vec{F}_2 = -(1N)\hat{i} - (2N)\hat{j}$$

This means (1N) along negative x-axis, (2N) along negative y-axis.

Hence vector 7 represents it.

$$\text{Net force } \vec{F} = 2\hat{i} - 6\hat{j}$$

$$\text{Magnitude of net force : } |\vec{F}|^2 = (2)^2 + (-6)^2 = 40$$

$$\therefore |\vec{F}| = \sqrt{40}N$$

10. Ans (2)

$$\tan \theta = m \Rightarrow \mu = \tan 60^\circ \Rightarrow \mu = \sqrt{3} = 1.732$$

11. Ans (4)

Parabola or straight line

12. Ans (1)

Change in momentum = Impulse

$$= \int F dt = \int_0^3 (12t + 3t^2) dt$$

$$= [6t^2 + t^3]_0^3 = 74 \text{ kg ms}^{-1}$$

13. Ans (2)

$$F = \frac{dP}{dt} = \text{Slope of P-t curve}$$

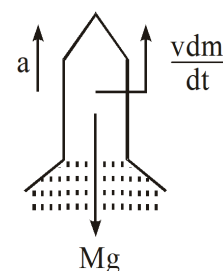
$$\text{So } \frac{F_A}{F_B} = \frac{\tan 30^\circ}{\tan 45^\circ} = \frac{1}{\sqrt{3}}$$

14. Ans (2)

$$F = m_1(4) = m_2(6) \Rightarrow m_1 = \frac{F}{4} \text{ \& } m_2 = \frac{F}{6}$$

$$a = \frac{F}{m_1 + m_2} = \frac{F}{\frac{F}{4} + \frac{F}{6}} = 2.4 \text{ m/s}^2$$

15. Ans (3)



$$\frac{v dm}{dt} - Mg = Ma$$

$$28 \times 10^2(x) = M(a + g)$$

$$= 25(2g)$$

$$= 490$$

$$\Rightarrow x = 0.175 \text{ kg/s}$$

Hence option (3)

16. Ans (4)

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$$\therefore \vec{F} = \frac{d\vec{p}}{dt} \text{ if } \vec{F} \neq 0 \Rightarrow \vec{p} \neq \text{constant}$$

and if work done by force is zero then kinetic energy remains constant.

17. Ans (4)

It may be positive, negative or zero.

18. Ans (1)

$$U = K(x^2 + y^2 + z^2)$$

$$\vec{F} = -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} - \frac{\partial U}{\partial z}\hat{k}$$

$$\vec{F} = -2K(x\hat{i} + y\hat{j} + z\hat{k})$$

19. Ans (3)

Theory Based

20. Ans (3)

$$KE = \frac{P^2}{2m}$$

$$KE \propto \frac{1}{m}$$

Less mass, more KE

21. Ans (3)

WET is applicable for all type of system either it contains one particle or many particles.

22. Ans (2)

$$U = 2x^4 - 27x$$

$$\frac{dU}{dx} = +8x^3 - 27$$

$$\frac{d^2U}{dx^2} = 24x^2 \rightarrow \text{minima}$$

$$\frac{d^2U}{dx^2} \text{ is +ve at } x = 3/2$$

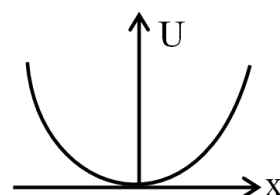
so stable equilibrium

23. Ans (1)

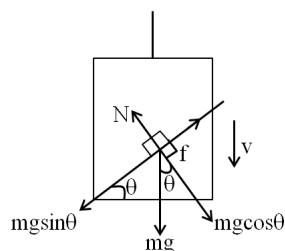
$$U = \frac{1}{2}kx^2$$

$$U \propto x^2$$

$\Rightarrow$ parabolic curve



24. Ans (3)



Block is at rest so

$$f = mg \sin \theta$$

$$W_f = |f| |s| \cos(90 + \theta)$$

$$W_f = (mg \sin \theta) (vt) (-\sin \theta)$$

$$W_f = -mg \sin^2 \theta (vt) \quad [\because s = vt]$$

$$W_f = -mg vt \sin^2 \theta$$

25. Ans (3)

$$p_1 = p \quad p_2 = p + \frac{200p}{100} = 3p$$

Percentage increase in kinetic energy

$$= \frac{E_2 - E_1}{E_1} \times 100 = \left[\frac{E_2}{E_1} - 1 \right] \times 100$$

$$= \left[\frac{p_2^2}{p_1^2} - 1 \right] \times 100 = \left[\frac{9p^2}{p^2} - 1 \right] \times 100$$

$$= 800\%$$

26. Ans (2)

$$\text{Energy loss} = \frac{1}{2}(0.5)(14)^2 - (0.5)(9.8)(9) \\ = 4.9 \text{ J}$$

27. Ans (1)

$$x = \frac{t^2}{2}$$

$$V = \frac{dx}{dt} = t$$

$$V(t=3) = 3 \text{ m/s}$$

$$W.D. = \frac{1}{2} m V_f^2 - \frac{1}{2} m V_i^2$$

$$W.D. = \frac{1}{2} \times 3 \times (3)^2 - 0 \\ = 13.5 \text{ J}$$

28. Ans (2)

From conservation of Mechanical energy

$$\frac{1}{2} M v^2 - M g \frac{L}{2} = - \frac{M}{2} \cdot g \cdot \frac{L}{4}$$

$$\frac{1}{2} M v^2 = \frac{3 M g L}{8}$$

$$v = \sqrt{\frac{3 g L}{4}}$$

29. Ans (1)

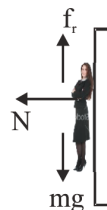
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$$w = \vec{F} \cdot \vec{s}$$

$$= 12 - 6 + 3x = 0$$

$$x = -2$$

30. Ans (3)



centripetal acceleration

31. Ans (1)

$$W = F s = \mu R s = \mu m g s$$

$$= 0.2 \times 50 \times 9.8 \times 1 = 98 \text{ J}$$

32. Ans (2)

The area of force-displacement graph gives work.

Work from $x = 0$ to $x = 25 \text{ m}$

$$W = \text{area (1)} + \text{area (2)} + \text{area (3)} + \text{area (4)}$$

$$= \frac{1}{2} \times 5 \times 4 + 4 \times 5 + \left(\frac{4+10}{2} \right) \times 5 + \frac{1}{2} \times 10 \times 10$$

$$= 10 + 20 + 35 + 50 = 115 \text{ J}$$

33. Ans (2)

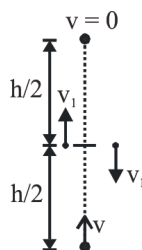
$$\frac{P_1}{P_2} = \frac{mgh}{t_1} \times \frac{t_2}{mgh} = \frac{t_2}{t_1} = \frac{20}{15}$$

34. Ans (2)

$$v = \sqrt{2gh}$$

$$v_1 = \sqrt{2g \frac{h}{2}} = \sqrt{gh} = \frac{v}{\sqrt{2}}$$

$$k = \frac{1}{2} m \left(\frac{v}{\sqrt{2}} \right)^2 = \frac{E}{2}$$



35. Ans (3)

$$\tan \theta = \frac{v^2}{Rg}$$

36. Ans (3)

For $v \geq \sqrt{5gl}$,

the bob will complete a vertical circular path

For $v_0 = \sqrt{5gl}$

At lowest point, $T - mg = \frac{m}{\ell}(5gl)$, $T = 6mg$

For $v_0 = 3\sqrt{gl}$

At highest point, $v^2 = v_0^2 - 4gl = 5gl$

$T + mg = \frac{m}{\ell}(5gl) \Rightarrow T = 4mg$

For $\sqrt{2gl} < v < \sqrt{5gl}$, the bob will execute projectile motion

For $v < \sqrt{2gl}$, the bob oscillates

37. Ans (3)

Ball will fall freely under gravity.

38. Ans (2)

$$\theta = \frac{(\omega_i + \omega_f) t}{2}$$

$$2\pi N = \frac{(20\pi + 40\pi)}{2} 10$$

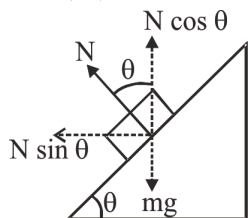
$$N = 150 \text{ revolution}$$

39. Ans (3)

Uniform motion $a_t = 0$ and $a_c = \omega^2 r = v\omega$

So $a_{net} = a_{cp} = v\omega$

40. Ans (1)



$$N \cos \theta = mg \quad \dots(1)$$

$$N \sin \theta = F_{cp} = \frac{mv^2}{r} \quad \dots(2)$$

$$\tan \theta = \frac{v^2}{rg} \Rightarrow \theta = \tan^{-1} \frac{v^2}{rg}$$

41. Ans (2)

$$w = \int \vec{F} \cdot d\vec{s}$$

$$P = \frac{dw}{dt}$$

$$\vec{F} = \frac{dp}{dt}$$

$$\vec{I} = \int \vec{F} \cdot dt = \Delta \vec{p}$$

42. Ans (1)

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Apply come from A to B

$$\frac{1}{2}m(9gR) + 0 = \frac{1}{2}mV^2 + mgR$$

$$\Rightarrow V^2 = 7gR$$

$$V = \sqrt{7gR}$$

43. Ans (4)

$$\omega = a - bt$$

$$\vec{\alpha} = \frac{d\vec{\omega}}{dt} = -b$$

$$a_t = \alpha r = -bR$$

$$\text{At } t = \frac{2a}{b}, \omega = a - b \left(\frac{2a}{b} \right)$$

$$\omega = -a$$

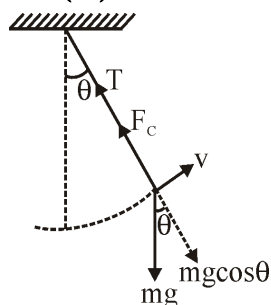
$$a_c = \omega^2 r$$

$$a_c = a^2 R$$

$$a_{net} = \sqrt{a_c^2 + a_t^2}$$

$$= \sqrt{(a^2 R)^2 + (bR)^2} = R^2 \sqrt{a^4 + b^2}$$

44. Ans (3)



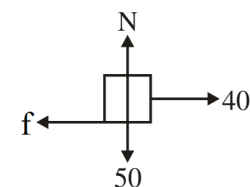
At angular position θ with the vertical

$$F_c = T - mg \cos \theta = \frac{mv^2}{\ell} \Rightarrow T = \frac{mv^2}{\ell} + mg \cos \theta$$

At $\theta_1 = 30^\circ$ and $\theta_2 = 60^\circ \Rightarrow v_1 > v_2$ and $\cos \theta_1 > \cos \theta_2$

$$\Rightarrow T_1 > T_2$$

45. Ans (2)



$$N = 50 \text{ N}$$

$$f_L = 0.6 \times 50 = 30 \text{ N}$$

$$f_K = 0.5 \times 50 = 25 \text{ N}$$

$$a = \frac{40 - 25}{5} = 3 \text{ m/s}^2$$

46. Ans (3)

Explanation

1. Diamagnetic vs. Paramagnetic:

- Diamagnetic molecules have **all electrons paired**.
- Paramagnetic molecules have **unpaired electrons**.

2. Octet Rule:

- Atoms achieve a stable electronic configuration by having 8 electrons in their valence shell. Molecules that strictly follow the octet rule do not have any expanded or incomplete octets.

Concept

1. NO₂ (Nitrogen Dioxide):

- Molecular structure: N is bonded to two O atoms with a bent geometry.
- Nitrogen has **one unpaired electron** (odd-electron molecule).
- It is **paramagnetic** and does **not** strictly follow the octet rule because of the unpaired electron.
- **NOT the answer.**

2. ClO₂ (Chlorine Dioxide):

- Molecular structure: Cl is bonded to two O atoms with a bent geometry.
- Chlorine has **one unpaired electron** (odd-electron molecule).
- It is **paramagnetic** and does **not** strictly follow the octet rule because of the unpaired electron.
- **NOT the answer.**

3. CO₂ (Carbon Dioxide):

- Molecular structure: Linear geometry with double bonds between carbon and two oxygen atoms.
- Carbon and oxygen atoms have **all electrons paired**.
- It is **diamagnetic** and strictly follows the **octet rule** (carbon has 4 bonds, and each oxygen has 8 valence electrons).
- **CORRECT ANSWER.**

4. NO (Nitric Oxide):

- Molecular structure: Linear geometry with a single bond between N and O.
- Nitrogen has **one unpaired electron** (odd-electron molecule).
- It is **paramagnetic** and does **not** strictly follow the octet rule because of the unpaired electron.
- **NOT the answer.**

Answer option 3

49. Ans (4)

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Radius $\propto$ number of shell

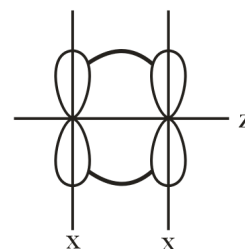
$$\text{radius} \propto \frac{1}{Z_{\text{eff}}/\text{effective nuclear}}$$

60. Ans (2)

NO₂

$$7 + 2 \times 8 = 23 e^-$$

62. Ans (4)



67. Ans (4)

Intermolecular H-bond $\propto$ Mpt

$\propto$ Viscosity

$$\propto \frac{1}{\text{Volatile nature}}$$

94. Ans (4)

NCERT-XII, Page # 66

95. Ans (4)

NCERT-XII Pg#106

99. Ans (4)

In all the given types of inheritance, the F₁ hybrid phenotype does not resemble either parent:

1. Incomplete Dominance - The F₁ hybrid shows an intermediate phenotype between both parents.

(Example: Pink flowers in *Mirabilis jalapa* from red and white parents.).

2. Codominance - Both parental traits appear simultaneously in the F₁ hybrid. (Example: AB blood group where both A and B alleles are expressed equally.).

3. Polygenic Inheritance - The trait is controlled by multiple genes, resulting in a blended phenotype that does not exactly match either parent. (Example: Human skin color and height.).

102. **Ans (1)**
NCERT Page No. 62
104. **Ans (3)**
NCERT Pg. # 62
105. **Ans (2)**
NCERT XII Pg. # 61
107. **Ans (2)**
The Correct answer is : 2. 3.1 %
 $\frac{1}{32} \times 100 =$
108. **Ans (2)**
NCERT pg no.73
110. **Ans (2)**
NCERT (XII) Pg # 122
111. **Ans (3)**
NCERT (XII) Pg. # (E)-91,92,93 (H)-101,102,103
113. **Ans (4)**
NCERT Pg. # 118, 119
115. **Ans (4)**
NCERT XII, Pg. # 101
116. **Ans (2)**
NCERT-XII, Pg. # 94
117. **Ans (4)**
NCERT (XII) Pg. # 92
118. **Ans (4)**
NCERT-XII (E), Pg#89,90
NCERT-XII (H), Pg#98,99
119. **Ans (2)**
NCERT XII Pg. # 103, line 2
120. **Ans (1)**
NCERT-XII Pg. No. # 110 Para- 6.5.3
123. **Ans (2)**
NCERT (XIIth) Pg. # 97
124. **Ans (1)**
Continuous

125. **Ans (2)** Telegram: @NEETxNOAH
Experiment based
 $2^4 = 16$, but in 4th generation
 $14 \text{ N}^{14} \text{ N}^{14} \text{ } 2 \rightarrow \text{N}^{15} \text{ N}^{14}$.
- 
126. **Ans (1)**
NCERT Pg. # 86, 87(E), 94, 96(H)
127. **Ans (3)**
NCERT XII, Pg # 83
130. **Ans (4)**
NCERT (XIIth) Pg. # 76
131. **Ans (4)**
NCERT-XII, Pg. # 80
133. **Ans (2)**
NCERT Pg # 94
134. **Ans (1)**
NCERT XII, pg.# 101
144. **Ans (3)**
NCERT(XII) Page No. # 135/148(H) Para:7.6
146. **Ans (4)**
NCERT Page # 132, 133
153. **Ans (4)**
The correct answer is 4. Both (1) and (2)
Explanation:
- Worked in Malay Archipelago: Alfred Wallace conducted extensive research in the Malay Archipelago, a vast region in Southeast Asia, where he made significant observations that led to his development of the theory of evolution by natural selection.
 - Had conclusions similar to those of Charles Darwin: Wallace independently formulated a theory of evolution that was remarkably similar to Darwin's. Both scientists recognized the importance of natural selection as the driving force behind the diversity of life.

157. Ans (1)

NCERT (XII) Pg# 128/139 (H) Para : 7.2

158. Ans (2)

Homo habilis probably was the first prehistoric man.

159. Ans (4)

Explanation :

Artificial selection (also known as selective breeding) is the process by which humans intentionally breed animals (or plants) with desirable traits in order to produce offspring with those traits. This is how new breeds of domestic dogs are developed. For example, breeders might select dogs with specific physical features or temperaments to create a

- new breed or improve an existing one.

Why the other options are incorrect :

1. **Sexual selection** : This is a form of natural selection where traits are selected based on their appeal to the opposite sex, not typically used for breeding domestic dogs in the way we see in artificial selection.
2. **Natural selection** : This occurs in nature without human intervention and favors traits that increase an organism's chances of survival and reproduction in the wild. However, it is not the mechanism behind the creation of new dog breeds.
3. **Youth selection** : This is not a recognized term in evolutionary biology or breeding and does not relate to the development of dog breeds.

Conclusion : New breeds in domestic dogs are developed through artificial selection, so the correct answer is

Option 4 : Artificial selection.

162. Ans (1)

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The correct answer is 1. Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

Assertion: When we see stars, we apparently are peeping into the past.

- This statement is true. Light travels at a finite speed (approximately 299,792,458 meters per second). Since stars are incredibly far away, the light we see from them has taken a significant amount of time to reach us. This means we are essentially seeing the star as it was in the past, not as it is in the present moment.

Reason: The light emitted from stars started its journey millions of years back and from trillions of kilometers away and reaching our eyes now.

- This statement is true and correctly explains the Assertion. The vast distances between stars and Earth mean that the light we observe has traveled for a long time, potentially millions of years. This time delay is the reason why we are essentially looking back in time when we observe stars.

164. Ans (3)

NCERT Pg. # 131-132

165. Ans (2)

NCERT Pg. # 127

166. Ans (2)

NCERT, Pg # 138(E), 151(H)

168. Ans (4)

NCERT Pg # 128/129

169. Ans (4)

NCERT, Pg # 137,138

170. **Ans (2)**
NCERT Pg # 128-129

171. **Ans (2)**
NCERT Pg # 129

172. **Ans (1)**
NCERT Pg # 132

174. **Ans (1)**
NCERT Pg. # 116

175. **Ans (1)** **Telegram: @NEETxNOAH**
NCERT-XII, Pg. # 102

176. **Ans (3)**
Ncert Pg No : 136, 137

180. **Ans (2)**
NCERT-XII Pg. # 136